IMovie – Project Design Phase

Date: 25 October 2026 · **Team ID:** [add team ID] · **Project Name:** IMovie – Movie Ticket Booking Application

1. Problem–Solution Fit

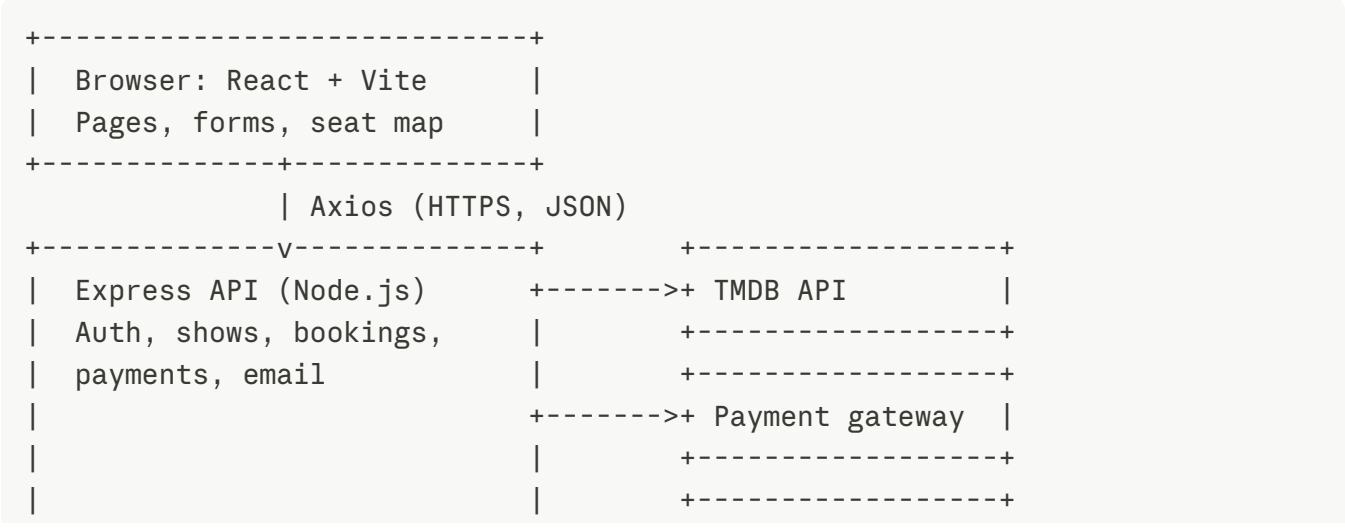
Element	Description
Customer segments	Moviegoers (students, working professionals, families) and theatre admins or managers
Customer problem	Queues at counters, no easy view of shows and seat availability, manual show management for theatres
Existing alternatives	Buying tickets at the counter, calling the theatre, large booking platforms where small theatres may not be listed
Triggers to act	A movie release, a weekend plan, wanting a guaranteed seat
Our solution	A web app where users find a show, select seats on a seat map, pay online and get an email confirmation; admins manage theatres and shows from a dashboard
Channels	Web application on desktop and mobile browsers
Emotions before	Frustration, uncertainty about seat availability
Emotions after	Confidence, convenience, saved time

2. Proposed Solution

Parameter	Description
Problem statement	Moviegoers need a quick way to book seats without queues; theatres need a simple tool to publish shows and see bookings
Idea and solution description	MERN web application with movie browsing (TMDB), admin theatre and show management, tier-based seat map, online payment and email confirmation
Novelty	Admin-defined seat tiers (balcony, middle, lower) with separate prices and row layout, so any theatre can describe its own hall
Social impact	Less waiting time for users, easier online presence for small theatres
Business model (proposed)	Small convenience fee per ticket, or a subscription for theatres
Scalability	Stateless REST API, indexed MongoDB collections, deployable on managed cloud services

3. Solution Architecture

3.1 Architecture overview



```
|                             +------->+ Email service    |
+--------------+--------------+        +------------------+
               | Mongoose
+--------------v--------------+
|  MongoDB (moviebook)        |
|  users, movies, theaters,   |
|  shows, bookings            |
+-----------------------------+
```

The TMDB call currently runs from the browser. It is planned to move behind the backend so the token is not exposed.

3.2 Components

Component	Responsibility	Status
React frontend	Home, movies, login, sign-up, admin dashboard and forms	Partly built
Express API	Theatre creation today; auth, shows, bookings, payments to be added	Partly built
MongoDB models	Movie, Theater, Show, User, Admin; Booking to be added	Partly built
Auth module	Password hashing, JWT, role checks, Google OAuth	Planned
Booking module	Seat map, seat hold, booking records	Planned
Payment module	Order creation, signature verification, webhook	Planned
Email module	Confirmation email after verified payment	Planned

3.3 Data relationships

- A **Theater** has many **Shows**; each Show is for one movie (TMDB id) at one date and time.
- A **User** has many **Bookings**; a Booking belongs to one Show and lists the selected seats with their tier and price.
- An **Admin** owns one or more Theaters.

3.4 Booking flow (target design)

1. User opens a movie and picks a show.
2. Frontend requests the seat map for that show; the API builds it from the theatre layout and existing bookings.
3. User selects seats; the API creates a pending booking and holds the seats for about 10 minutes.
4. The API creates a payment order and the frontend opens the payment window.
5. After payment, the API verifies the signature, marks the booking confirmed and releases nothing.
6. The API sends the confirmation email. If payment fails or the hold expires, the seats are released.

3.5 Key design decisions

Decision	Reason
Store seat tiers as layout (rows and seats per row) plus tier price	Lets each theatre define its own hall without a fixed template
Hold seats before payment	Prevents two users paying for the same seat
Verify payments on the server	The browser result cannot be trusted
JWT with roles	Keeps the API stateless and protects admin routes
Send email only after verified payment	Users never get a confirmation for an unpaid booking